

CSCE 670 INFORMATION RETRIEVAL · MODULE 12 · LECTURE NOTES

Exposure and Consequences

What a ranking does to the things being ranked.

MODULE OUTCOMES

- Explain exposure and why rank position is the resource being allocated.
- Explain feedback loops in deployed rankers.
- Explain the fairness formulations and their disagreements.
- Explain filter bubbles honestly, including the evidence.
- Assess a ranking system's consequences.

1 Exposure

ATTENTION FALLS STEEPLY WITH RANK, SO A RANKING ALLOCATES A SCARCE RESOURCE

Click-through rate falls sharply with position (Module 05 §4's position bias) — so **rank 1 receives many times the attention of rank 10, and rank 11, below the fold, receives almost none at all.**

Which means a ranking is not a neutral ordering but an allocation of attention — and **attention is revenue, reach, and opportunity for whoever produced the document**, which makes the ranking a consequential decision about people rather than a presentation choice.

And the allocation is extremely unequal: a very small difference in relevance score produces a very large difference in exposure, because the position-to-attention curve is so steep near the top.

So two documents of nearly equal relevance receive wildly unequal outcomes — which is a property of the presentation rather than of the ranking function, and is where the fairness question starts (§3): nothing in the scoring is unfair, and the outcome is.

- **Ties should be broken randomly, or rotated across impressions**, because a **deterministic tiebreak gives one document all of the exposure permanently** — and document id order is a common and entirely arbitrary tiebreak.
- **Randomised ranking spreads exposure** at a small quality cost — **and it also produces unbiased click data, which is a second and substantial benefit** (Module 05 §4, and §2's loop).
- **Exposure can be measured and reported**: the distribution of impressions across the catalogue, **which is CSCE 676 Module 09 §4's coverage metric** arriving here as a fairness measurement.
- **And it can be constrained**, by an explicit amortised exposure target across queries — **which is a policy decision rather than a technical one** and should be made and stated as such.

- **But all of it costs relevance**, which has to be stated rather than hidden — there is no free fairness intervention here. **Randomisation buys two things at once** — fairer exposure and unbiased training data — **which makes it a considerably better deal than it first appears** and is the intervention to reach for first.

2 Feedback

YOUR TRAINING DATA IS THE OUTPUT OF YOUR PREVIOUS RANKER

Clicks are only ever observed on documents you actually showed — so **the data records what you ranked highly rather than what was best** (CSCE 676 Module 09 §3's feedback loop, in its original setting).

So a document ranked low receives no clicks, therefore looks unpopular, therefore stays ranked low — **a self-fulfilling loop that no amount of model improvement escapes**, because the model is learning from data the previous model generated.

Which means the system converges toward its own early judgements, and **a document's initial placement matters more than its actual quality** — which is a striking and uncomfortable property of a system that looks purely meritocratic.

And the corrections are all forms of exploration: randomisation (§1), inverse propensity weighting of the click data, and deliberate exposure of new items — **all of which cost short-term relevance and buy long-term correctness**, which is a trade worth making explicitly rather than by default.

3 Fairness

Criterion	What it requires	The tension
Demographic parity of exposure	Groups receive exposure proportional to their share of the catalogue.	It ignores genuine relevance differences, so it can require showing worse results.
Exposure proportional to relevance	Documents of equal relevance receive equal exposure.	It requires relevance to have been measured fairly, which §2's loop undermines.
Individual fairness	Similar documents are ranked similarly.	It requires a similarity metric that nobody has, and the metric choice carries the whole decision.
Amortised fairness	Equal exposure across many queries rather than within one result list.	Practical and implementable, and it needs a session or time window concept to amortise over.

These criteria are mutually incompatible in general — satisfying one can require violating another, which is a formal result rather than an engineering difficulty — **so choosing a fairness criterion is a value judgement and cannot be derived from the data. That is the honest position, and the incompatibility is the point: it should not be allowed to become a technical question with a technical answer, because it does not have one.**

AND THE TWO-SIDED MARKET MAKES IT HARDER

A ranking serves searchers and simultaneously allocates outcomes to document providers — sellers, publishers, authors, creators — **and the two sides have different interests, which a single relevance objective does not represent at all.**

So 'optimal' is underdetermined: the ranking that is best for searchers in aggregate may starve a provider whose content is good and only slightly less popular, which **degrades the catalogue over time** as that provider leaves.

Which connects directly to §2: provider starvation is itself a feedback loop, and the catalogue the system depends on erodes — the same structure as Module 11 §4's collective-action problem, at a different level.

So the honest framing is a multi-objective one with the weights stated — **rather than a single relevance objective with all the other effects left unmeasured**, which is the default and is the thing this module asks you to notice.

4 Filter bubbles, and assessment

THE FILTER BUBBLE ARGUMENT IS PLAUSIBLE AND THE EVIDENCE IS MORE MIXED THAN THE DISCUSSION SUGGESTS

The mechanism is real: personalisation shows you more of what you previously engaged with, which narrows the range of what you see, and §2's feedback loop reinforces it at the system level.

And the measured effects are smaller and more complicated than the argument predicts: several careful studies find that algorithmic feeds expose people to *more* diverse sources than their own unaided choices would, because self-selection is also strongly narrowing and the comparison baseline matters.

So the honest position is that both the mechanism and the counter-evidence are real — and that the effect depends heavily on the platform, the domain, the measure of diversity, and the comparison chosen, which is precisely why the literature disagrees rather than one side being mistaken.

What follows practically is measurable regardless of how the debate resolves: measure the diversity of what you serve, report it, and give users control over personalisation — which is defensible on its own terms and does not require settling the empirical question first.

- The exposure distribution across the catalogue, reported rather than assumed uniform — which is one query against your impression logs.
- Whether new items can ever surface, which tests the feedback loop directly (§2) and is answerable by tracking a cohort of newly added documents.
- The diversity of a typical result list, and separately the diversity of what a typical user sees over time — which are different questions and §4's debate is about the second.
- Which fairness criterion you chose and why — stated as a value judgement rather than presented as derived (§3), because presenting a value judgement as a technical result is the characteristic dishonesty in this area.
- And who is affected besides the searcher, which is the question that is usually not asked at all. 'Who is affected besides the searcher' is the question this module exists to make askable — and it has an answer in every deployed ranking system, whether or not anybody has looked for it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Singh & Joachims — Fairness of Exposure in Rankings (free)	§1 and §3 formalised — exposure as the allocated resource, with the optimisation framework.
Biega, Gummadi & Weikum — Equity of Attention (free)	§3's fourth row — amortised fairness, which is the most implementable of the criteria.

Resource	What it covers for this module
Joachims et al. — Unbiased Learning-to-Rank with Biased Feedback (free)	§2's correction — inverse propensity weighting, and what it requires to work.
Bakshy, Messing & Adamic, and the subsequent debate (free)	§4's counter-evidence and its critics — read the paper and at least one critique, since the disagreement is the lesson.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Measure click-through by position** on any log you have, and plot the curve.
- 2 **Compute the exposure a one-position improvement buys.**
- 3 **Find a deterministic tiebreak** in your system and make it random.
- 4 **Measure the exposure distribution** across your catalogue.
- 5 **Track a cohort of new documents** and report whether any surfaced.
- 6 **Simulate the feedback loop:** train on your own rankings' clicks for several rounds and measure coverage.
- 7 **Implement two fairness criteria** and show a case where they disagree.
- 8 **Measure the relevance cost** of your chosen criterion.
- 9 **Measure result-list diversity** and per-user diversity over time.
- 10 **List who besides the searcher** is affected by your ranking, and how.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is a ranking an allocation rather than an ordering?
- 2 Why do near-equal documents get unequal outcomes?
- 3 Give four consequences of the exposure curve, and what randomisation buys.
- 4 State the feedback loop and why model improvement does not escape it.
- 5 Name three corrections and what they cost.
- 6 Name four fairness criteria and the tension in each.
- 7 Why can a fairness criterion not be derived from data?
- 8 Why does a two-sided market make 'optimal' underdetermined?
- 9 State the filter bubble position honestly, with both sides.

10 Give five things to assess, and the question this module makes askable.